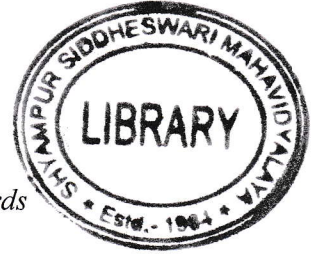
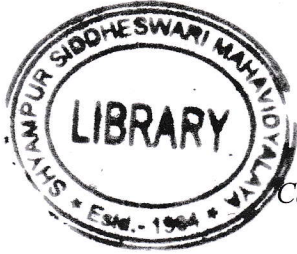


2022

COMPUTER SCIENCE — GENERAL

Paper : GE/CC-3

Full Marks : 50



The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **question no. 1** and **any four** questions from the rest.

I. Answer **any five** questions :

2×5

- (a) Differentiate between memory mapped I/O and I/O mapped I/O.
 - (b) What are the major advantages of EEPROM over EPROM and PROM?
 - (c) State the role of stack pointer.
 - (d) State the role of Program Counter (PC).
 - (e) Write down the functionality of program status word.
 - (f) What is Virtual Memory?
2. Write down the flowchart/algorithm for Booth's Multiplication. Perform Booth's multiplication for the following numbers : $(3)_{10} \times (-7)_{10}$. 5+5
3. (a) Write the differences between RISC and CISC.
(b) Perform the following subtraction using the 2's complement method : $(1001)_2 - (1111)_2$. 5+5
4. (a) Write down the significances of Polling.
(b) In how many ways DMA controller transfers the data? Define each of them. 4+(1+5)
5. (a) Write short notes on :
(i) Indirect addressing mode
(ii) Direct addressing mode.
(b) Write the functions of PLA. (3+3)+4
6. (a) Write down the key features of SCSI and PCI bus.
(b) Explain any one of the bus arbitration method. 6+4
7. (a) Describe the working principle of VDU.
(b) What are the ways a negative integer number can be represented? 6+4
8. (a) Write down the characteristics of Hardwired Control and Microprogrammed Control unit.
(b) Define Instruction Cycle. Explain its operation with a flowchart. 3+(2+5)
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